

NU-87

BLE MCU Module

- Model No. NCWB87R01VC / NCWB87R01VI
- Dual-band Wi-Fi & BLE Combo NUCODE NU Series



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1. FUNCTIONAL DESCRIPTION

1.1. BLOCK DIAGRAM

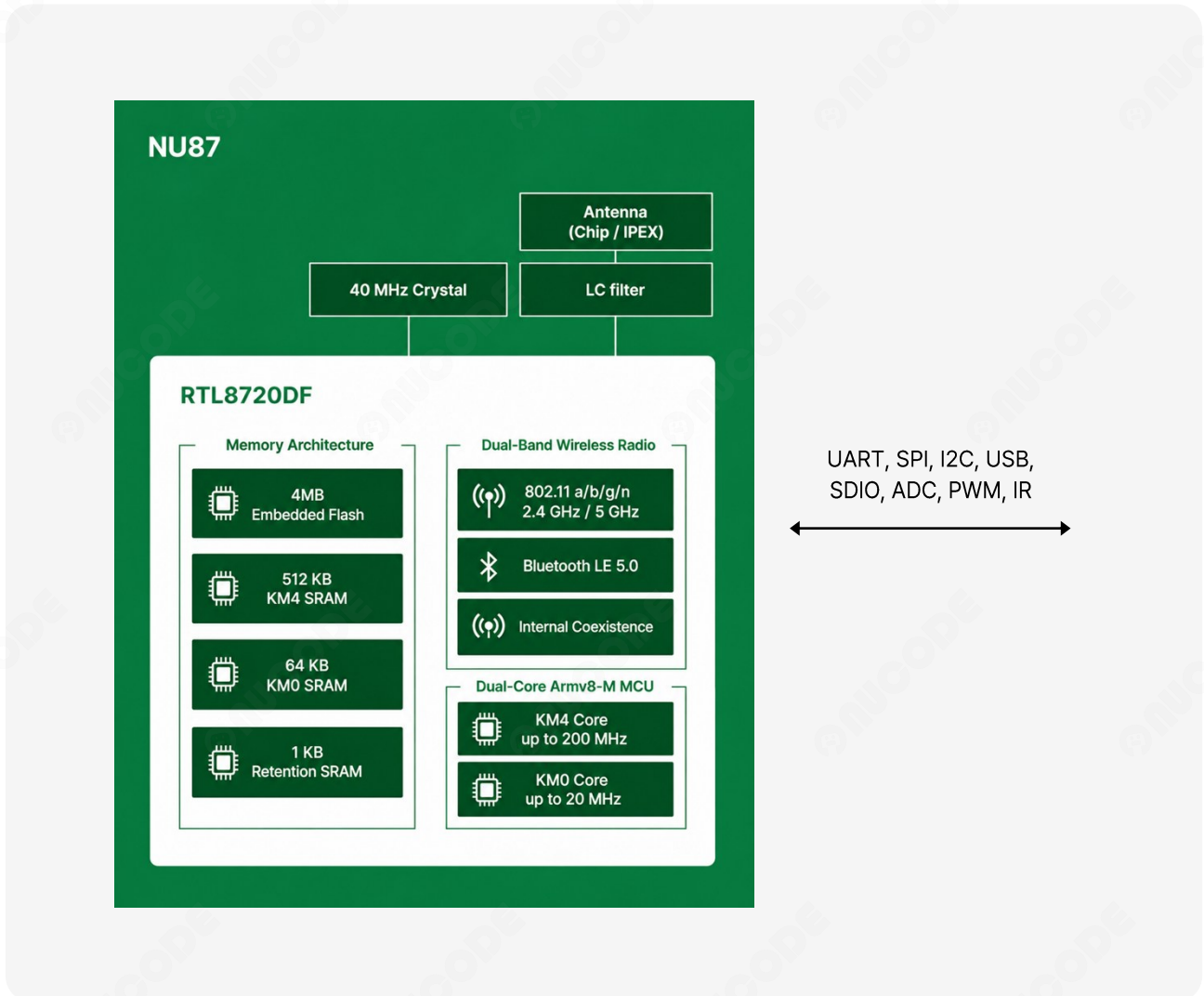


Figure 1-1 NU-87 Block diagram

1.2. FEATURES

1.2.1. CPU

RTL8720DF with embedded 4 MB Flash, dual-core Armv8-M MCU

KM4(Cortex-M33 compatible) up to 200 MHz, KM0(Cortex-M23 compatible) up to 20 MHz

- KM4 SRAM: 512 KB
- KM0 SRAM: 64 KB
- Retention SRAM: 1 KB

1.2.2. Wi-Fi®

IEEE 802.11 a/b/g/n (Wi-Fi® 4 compatible), 1x1 dual-band Wi-Fi® (2.4 GHz / 5 GHz)

- Operating frequency: 2412~2484 MHz, 5180~5825 MHz

- Supports 20 MHz / 40 MHz bandwidth, up to MCS7

1.2.3. Bluetooth® LE

Bluetooth® Low Energy 5.0

- Operating frequency: 2402~2480 MHz
- Supports central and peripheral modes
- Internal Wi-Fi®/Bluetooth® coexistence for shared antenna operation

1.2.4. Power Management

- Supports active, sleep and deep-sleep modes
- All SRAM retained in sleep mode
- 1 KB retention SRAM available in deep-sleep mode
- Low-power operation with retained I/O support

1.2.5. Security

- AES/DES/SHA Integrated hardware engine
- Arm TrustZone-M
- Secure boot support
- Debug port protection
- Secure eFuse support

1.2.6. Peripherals

- UART, SPI, I2C®, SD/SDIO, DMIC(Digital Microphone Interface), USB 2.0, I2S®, PWM, IR, ADC

1.3. APPLICATIONS

NU-87 is available in a variety of applications, including:

- Smart Home Devices
- Home Appliances
- Industrial Automation
- IoT Gateways and Sensor Hubs
- Wireless Data Loggers
- Healthcare Devices
- Consumer Electronics
- Audio Devices
- Service Robots
- Wi-Fi® + BLE Bridge Devices

2. PIN DEFINITIONS

2.1. PIN LAYOUT

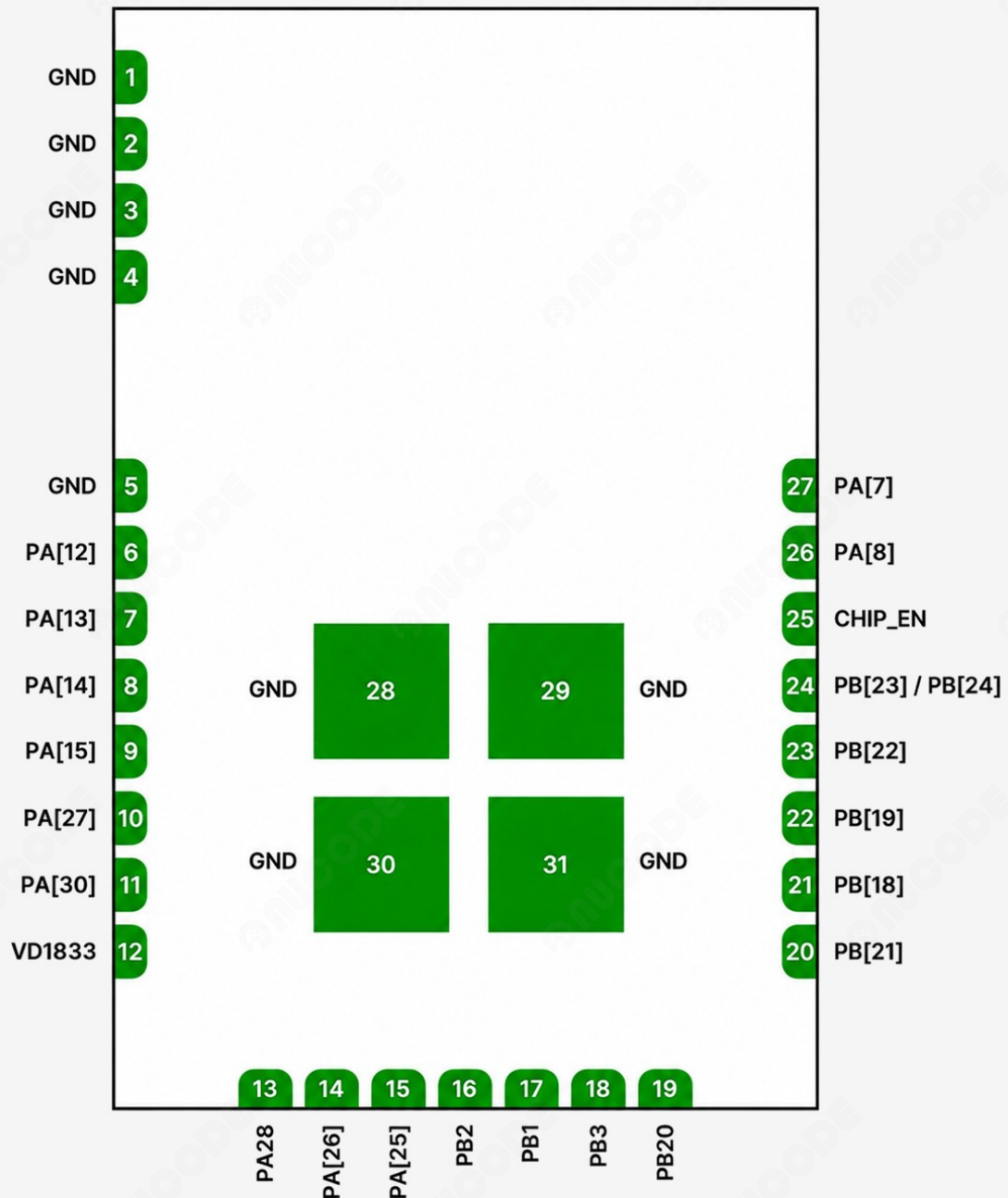


Figure 2-1. NU-87 Pin layout

2.2. PIN DESCRIPTION

No.	Name	Function	Description
1	GND	Ground	
2	GND	Ground	
3	GND	Ground	
4	GND	Ground	
5	GND	Ground	
6	PA[12]	GPIO, LP_UART_TXD, SPI1_MOSI, HS_PWM0, LP_PWM0, I2S_MCLK, ANT_SEL_N, GRANT_BT, EN_EXLNA, KEY_ROW0, LGPIO[0]	
7	PA[13]	GPIO, LP_UART_RXD, SPI1_MISO, HS_PWM, LP_PWM1, I2S_SD_TX1, ANT_SEL_P, GRANT_BT_N, EN_EXPA, KEY_ROW1, LGPIO[1]	
8	PA[14]	GPIO, LP_UART_RTS, SPI1_CLK, I2S_SD_TX2, RTC_OUT, ANT_SEL_N, BT_DIS, KEY_ROW2, LGPIO[2]	
9	PA[15]	GPIO, LP_UART_CTS, SPI1_CS, RTC_EXT_32K, ANT_SEL_P, BT_WAKE_HOST, KEY_ROW3, KEY_COL6, LGPIO[3]	
10	PA[27]	GPIO, LP_UART_RTS, SWD_DATA, WLAN_ACT, Boot strap	NORMAL_MODE_SEL - 1: Normal operation mode - 0: Enter test/debug mode
11	PA[30]	GPIO, HS_SPI_CLK, HS_PWM7, LP_PWM1, EXTBT_UART_RTS, Boot strap	SPS_SEL - 1: Internal 1.1V regulator works at SPS mode - 0: Internal 1.1V regulator works at LDO mode
12	VD1833	POWER	1.8V / 3.3V power supply pin
13	PA[28]	GPIO, LP_UART_CTS, HS_SPI_CS, HS_PWM6, LP_PWM0, RREF, BT_CK	12 kΩ / 1% resistor required between RREF and GND when using USB
14	PA[26]	GPIO, LP_UART_TXD, HS_SPI_MISO, IR_RX, LP_I2C_SDA, HS_PWM5, LP_PWM5, HSDP, BT_ACT, KEY_COL0, LGPIO[3], USB D+	Shared with LP_I2C SDA — USB and LP_I2C cannot be used simultaneously
15	PA[25]	GPIO, LP_UART_RXD, HS_SPI_MOSI, IR_TX, LP_I2C_SCL, HS_PWM4, LP_PWM4, HSDM, MBOX_I2C_INT, KEY_COL1, LGPIO[2], USB D-	Shared with LP_I2C SCL — USB and LP_I2C cannot be used simultaneously
16	PB[2]	GPIO, LP_UART_RXD, DMIC_DATA, SGPIO, ANT_SEL_P, PCM_CLK, EN_EXPA, HS_TIM5_TRIG, ADC_CH5	ADC input No pull resistor required when configured as ADC (NO PULL)
17	PB[1]	GPIO, LP_UART_TXD, DMIC_CLK, SGPIO_OUT, ANT_SEL_N, BT_STE, EN_EXLNA, HS_TIM4_TRIG, ADC_CH4	ADC input No pull resistor required when configured as ADC (NO PULL) Default function is SWD_CLK.
18	PB[3]	GPIO, SWD_CLK, PCM_SYNC, ADC_CH6	ADC input No pull resistor required when configured as ADC (NO PULL)

19	PB[20]	GPIO, HS_UART_TXD, HS_UART0_CTS, SPI0_CLK, SPI_DATA0, HS_I2C_SCL, SD_CMD, HS_PWM12, LP_PWM0, I2S_CLK, LCD_VSYNC_TE	
20	PB[21]	GPIO, HS_UART_RXD, HS_UART0_RTS, SPI0_CS, SPI_CLK, HS_I2C_SDA, SD_CLK, HS_PWM13, LP_PWM1, I2S_WS, LCD_RS, QDEC_IDX	
21	PB[18]	GPIO, HS_UART0_RXD, HS_UART_RTS, SPI0_MOSI, SPI_CS, SD_D2, HS_PWM10, LP_PWM4, SWD_CLK, LCD_D14	
22	PB[19]	GPIO, HS_UART0_TXD, HS_UART_CTS, SPI0_MISO, SPI_DATA1, SD_D3, HS_PWM11, LP_PWM5, SWD_DATA, I2S_SD_TX0, LCD_D15	
23	PB[22]	GPIO, IR_RX, SPI_DATA3, SD_D0, HS_PWM14, LP_PWM2, I2S_SD_RX, LP_TIM4_TRIG, LCD_RD, QDEC_PHB, SGPIO_OUT, EXTBT_UART_CTS	
24	PB[23] / PB[24]	PB23: GPIO, IR_TX, SPI_DATA2, SD_D1, HS_PWM15, LP_PWM3, I2S_MCLK, LP_TIM5_TRIG, LCD_WR, QDEC_PHA, SGPIO_OUT, EXT_32K PB24: GPIO, SD_WP, HS_PWM16, LP_PWM4, DMIC_CLK	Co-bonded pin — PB[23] and PB[24] cannot be used simultaneously
25	CHIP_EN	Chip Enable	Enable chip - 1: Enable chip - 0: Shut down chip
26	PA[8]	GPIO, LOGUART_RX, ANT_SEL_N	
27	PA[7]	GPIO, LOGUART_TX, ANT_SEL_P, Boot strap	UART_DOWNLOAD - 1: Boot from Flash - 0: Download image from UART
28	GND	Ground	
29	GND	Ground	
30	GND	Ground	
31	GND	Ground	

Table 2-1. Pin description

2.3. DEFAULT PULL DESCRIPTION

Default pull-up/down configuration	Port Name
Internal UP	PA[7], PA[8], PA[27]
EfusePullCtrl	PA[13], PA[15], PA[25], PA[28], PB[1], PB[19], PB[22]
External UP	PA[30]

Table 2-2. Default pull description

3. RESET AND DEBUG

3.1. BOOT MODES

The boot mode of NU-76 is determined by the power on states of dedicated boot strap pins shared with the GPIOs. During power on, the device samples the strap pin levels to select the boot mode. The relevant pins on NU-76 are UART_DOWNLOAD(PA[7]), NORMAL_MODE_SEL(PA[27]) and SPS_SEL(PA[30]).

NU-76 supports following boot-related modes:

- Normal Boot from Flash mode:
The module boots from the embedded Flash memory when UART_DOWNLOAD(PA[7]) is high. During normal boot operating, NORMAL_MODE_SEL(PA[27]) shall remain high.
- UART Download mode:
The module enters UART Download mode when UART_DOWNLOAD(PA[7]) is low during power on, instead of normal boot from Flash.
- Test/Debug mode:
The module enters Test/Debug mode when NORMAL_MODE_SEL(PA[27]) is low during power on and the normal boot will fail. This mode requires the pin must not be pulled low during power on.
- Regulator mode (SPS/LDO):
The internal 1.1V regulator mode selected with the state of SPS_SEL(PA[30]) during power on. SPS mode selected when SPS_SEL(PA[30]) is high, LDO mode selected when SPS_SEL(PA[30]) is low.
The NU-87 has an internal 10 kΩ pull-up resistor and operates in SPS mode by default.

In addition, CHIP_EN pin controls NU-76 enables or shutdown. Driving CHIP_EN high enables the module, driving CHIP_EN low shuts the module down.

3.2. STRAP PINS

The pins listed in table are shared with RTL8720DF Power on Trap Pins. These pins may be used as GPIO after boot, but their levels during power on affect selection of boot mode and must be controlled accordingly.

Symbol	Pin	Default configuration	Description
NORMAL_MODE_SEL	PA[27]	Internal pull-up	1: Normal operation mode 0: Enter test/debug mode
UART_DOWNLOAD	PA[7]	Internal pull-up	1: Boot from Flash 0: Download image from UART
SPS_SEL	PA[30]	External pull-up	1: Internal 1.1V regulator works at SPS mode 0: Internal 1.1V regulator works at LDO mode 10k pull-up resistor is mounted on the module
CHIP_EN	CHIP_EN	-	1: Enable chip 0: Shut down chip

Table 3-1. Strap pins

4. PERIPHERALS

4.1. DATA INTERFACES

This section describes the peripherals available on NU-87. Some functions can be used only on specific pin and simultaneous use of multiple data interfaces may cause pin conflicts. For pin assignments, refer to [2. Pin Definitions](#).

4.1.1. SPI

UART is an asynchronous serial interface consisting of transmit (TXD) and receive (RXD) signals. RTS/CTS handshake signals can optionally be used for flow control.

NU-76 provides 3 UART instances. HS_UART0 is a high-speed instance which supporting maximum 6 Mbps, LP_UART0 and LB_UART1 belong to the low-power domain. All three instances support DMA transfers. PA[7] and PA[8] are configured as LOGUART by default and are not recommended for general-purpose UART use.

Instance	TXD	RXD	RTS	CTS	Max Baud Rate
HS_UART0	PB[19]	PB[18]	PB[21]	PB[20]	6 Mbps
HS_UART	PB[20]	PB[21]	PB[18]	PB[19]	6 Mbps
LP_UART0	PA[12]	PA[13]	PA[14]	PA[15]	115200 bps
LP_UART1	PA[26] / PB[1]	PA[25] / PB[2]	PA[27]	PA[28]	115200 bps
LOGUART	PA[7]	PA[8]	-	-	-

Table 4-1. SPI instances

* Note:

- HS_UART0 and HS_UART share PB[18]~PB[21] and cannot be used simultaneously.
- LOGUART is used for bootloader communication at 1.5 Mbps. The baud rate may vary depending on firmware or application configuration.

4.1.2. SPI

SPI is a synchronous serial interface using 4 signals; clock (CLK), chip select (CS), master-out-slave-in (MOSI), and master-in-slave-out (MISO). It supports both master and slave modes.

NU-76 provides 2 SPI instances. SPI0 supports maximum 50 Mbps, SPI1 supports maximum 25 Mbps. Both instances are powered off in sleep and deep-sleep modes.

Instance	MOSI	MISO	CLK	CS	Max Clock	Mode
SPI0 Group1	PA[25]	PA[26]	PA[30]	PA[28]	50 MHz	Master / Slave
SPI0 Group2	PB[18]	PB[19]	PB[20]	PB[21]	50 MHz	Master / Slave
SPI1	PA[12]	PA[13]	PA[14]	PA[15]	25 MHz	Master

Table 4-2. SPI instance

* Note: SPI0 Group1 and Group2 cannot be used simultaneously.

4.1.3. I2C®

I2C® is a two-wire serial interface using only SDA and SCL signals. Both master and slave can be connected on a single bus, and it supports 7-bit & 10-bit addressing modes.

1 I2C® instance is available on NU-76 by PA25 & PA26. Two I2C® instances are available on NU-76. LP_I2C uses PA[25]/PA[26], which are shared with the USB interface (HSDM/HSDP) and cannot be used simultaneously with USB. HS_I2C uses PB[20]/PB[21], which are shared with SPI0 and SDIO.

Instance	SDA	SCL	Max Speed	Note
LP_I2C	PA[26]	PA[25]	400 kHz	Shared with USB (HSDM/HSDP)
HS_I2C	PB[21]	PB[20]	400 kHz	Shared with SPI0, SDIO

Table 4-3. I2C® instances

4.1.4. I2S®

I2S® is a serial bus standard for digital audio data transfer, consisting of bit-clock, word select and serial data signals. TX & RX can operate independently.

NU-76 provides 1 I2S® instance supporting mono, stereo and 5.1 channel configurations. In sleep & deep-sleep mode, the instance is powered off.

Signal	Pin	Description
I2S_CLK (BCLK)	PB[20]	Bit clock
I2S_WS	PB[21]	Word select
I2S_SD_RX	PB[22]	Serial data input
I2S_SD_TX0	PB[19]	Serial data output (ch0)
I2S_SD_TX1	PA[13]	Serial data output (ch1)
I2S_SD_TX2	PA[14]	Serial data output (ch2)
I2S_MCLK	PA[12] / PB[23]	Master clock (optional)

Table 4-4. I2S® instances

4.1.5. DMIC

DATA signals. It allows direct connection of digital microphones without an external analog microphone circuit.

Signal	Pin	Description
DMIC_CLK	PB[1] / PB[24]	Digital microphone clock
DMIC_DATA	PB[2]	Digital microphone data

Table 4-5. DMIC instance

4.1.6. USB

The USB 2.0 interface supports both host and device modes. It uses two differential signals, HSDM and HSDP, with 12k Ω (1%) resistor connected from the RREF pin to GND.

It supports all high-speed (480 Mbps), full-speed (12 Mbps) and low-speed (1.5 Mbps) modes. The interface is powered off in sleep & deep-sleep mode.

Note that PA25/PA26 cannot be used simultaneously with LP_I2C.

Signal	Pin	Description
HSDM	PA25	USB D- / I2C SDA (shared)
HSDP	PA26	USB D+ / I2C SCL (shared)
RREF	PA28	Analog reference (Connect 12k Ω /1% pull-down when using USB feature)

Table 4-6. USB instances

4.1.7. SDIO

SDIO is an interface for communicating with SD cards and SDIO devices using 4-bit data bus with clock frequencies maximum 50 MHz. It supports SDIO 2.0 SDR25 transfer mode (max 25 MB/s), SD card host mode and INIC mode (SDIO-to-Wi-Fi® bridge). PB[20] & PB[21] are shared with HS_I2C & SPIO, PB[18] & PB[19] are shared with SPIO & HS_UART0.

Signal	Pin	Description
SD_CMD	PB[20]	Command
SD_CLK	PB[21]	Clock
SD_D0	PB[22]	Data bit 0
SD_D1	PB[23]	Data bit 1
SD_D2	PB[18]	Data bit 2
SD_D3	PB[19]	Data bit 3
SD_WP	PB[24]	Write protect (optional)

Table 4-7. SDIO instances

4.2. CONTROL & SIGNAL INTERFACE

4.2.1. PWM

PWM controls the duty cycle of digital output signals and is used for motor control, LED dimming and waveform generation. It consists of a 16-bit auto-reload up-counter and 8-bit prescaler. NU-76 exposes 13 HS domain channels and 6 LP domain channels on external pins.

Channel	Pin	Domain
HS_PWM0	PA[12]	High-speed
HS_PWM1	PA[13]	High-speed
HS_PWM4	PA[25]	High-speed
HS_PWM5	PA[26]	High-speed
HS_PWM6	PA[28]	High-speed
HS_PWM7	PA[30]	High-speed
HS_PWM10	PB[18]	High-speed
HS_PWM11	PB[19]	High-speed
HS_PWM12	PB[20]	High-speed
HS_PWM13	PB[21]	High-speed
HS_PWM14	PB[22]	High-speed
HS_PWM15	PB[23]	High-speed
HS_PWM16	PB[24]	High-speed
LP_PWM0	PA[12] / PA[28] / PB[20]	Low-power
LP_PWM1	PA[13] / PA[30] / PB[21]	Low-power
LP_PWM2	PB[22]	Low-power
LP_PWM3	PB[23]	Low-power
LP_PWM4	PA[25] / PB[18] / PB[24]	Low-power
LP_PWM5	PA[26] / PB[19]	Low-power

Table 4-8. PWM instances

4.2.2. IR

The IR interface is used for transmitting and receiving infrared remote-control signals. The carrier frequency is configurable from 25 kHz to 500 kHz. 1 TX and 1 RX channel are provided, each with a 32×4-byte FIFO.

Signal	Pin	Description
IR_TX	PA[25] / PB[23]	IR transmit
IR_RX	PA[26] / PB[22]	IR receive

Table 4-9. IR instances

4.3. ANALOG INTERFACES

4.3.1. ADC

The ADC features a 12-bit SAR architecture and supports an input range from 0 V to VDD1833. 3 ADC channels are exposed on NU-76. The ADC supports a maximum conversion rate of 1 MSPS. The ADC remains active in sleep and deep sleep modes and can be used as a wake source.

Channel	Pin	Input Range	Resolution
ADC4	PB[1]	0 ~ 3.3 V	12-bit
ADC5	PB[2]	0 ~ 3.3 V	12-bit
ADC6	PB[3]	0 ~ 3.3 V	12-bit

Table 4-10. ADC instances

5. ELECTRICAL CHARACTERISTICS

5.1. ABSOLUTE MAXIMUM RATINGS

Maximum ratings are the extreme limits to which the chip can be exposed for a limited amount of time without permanently damaging it. Exposure to absolute maximum ratings for prolonged periods of time may affect the reliability of the device.

Symbol	Parameter	Min.	Max.	Unit
VD1833	Module supply voltage	-0.3	3.63	V
V _{IN}	Input voltage on PAx & PBx pins	-0.3	VD1833+0.3	V
T _{STORE}	Storage temperature range	-65	+150	°C
MSL	Moisture Sensitivity Level		MSL3	
HBM	ESD Human Body Model		Class 2	
CDM	ESD Charged Device Model		Class C2	

Table 5-1. Absolute maximum ratings

5.2. RECOMMENDED OPERATING CONDITIONS

The operating conditions are the physical parameters that the chip can operate within.

Symbol	Parameter	Min.	Typ.	Max.	Unit
VD1833	Module supply voltage	1.76	1.8/3.3	3.63	V
I _{VDD33}	3.3V rating current	-	-	450	mA
I _{VDD18}	1.8V rating current	-	-	800	mA
T _A	Operating temperature	-20	-	85	°C

Table 5-2. Recommended operating conditions

5.3. DC CHARACTERISTICS

The DC Characteristics define the voltage levels and current parameters of digital I/O pins. The specifications vary depending on the module supply voltage. All values are specified at the recommended operating conditions unless otherwise noted.

Symbol	Parameter	Min.	Typ.	Max.	Unit
V_{IH}	Input high voltage	2.0	-	-	V
V_{IL}	Input low voltage	-	-	0.8	V
V_{OH}	Output high voltage	2.4	-	-	V
V_{OL}	Output low voltage	-	-	0.4	V
I_{IL}	Input leakage current	-10	± 1	10	μA
C_{IN}	Input capacitance	-	3.7	-	pF

Table 5-3. Digital I/O pin DC Characteristics (VD1833 = 3.3V)

Symbol	Parameter	Min.	Typ.	Max.	Unit
V_{IH}	Input high voltage	$0.65 \times VDD$	-	-	V
V_{IL}	Input low voltage	-	-	$0.35 \times VDD$	V
V_{OH}	Output high voltage	$VDD - 0.45$	-	-	V
V_{OL}	Output low voltage	-	-	0.45	V
I_{IL}	Input leakage current	-10	± 1	10	μA
C_{IN}	Input capacitance	-	3.7	-	pF

Table 5-4. Digital I/O pin DC Characteristics (VD1833 = 1.8V)

5.4. POWER CONSUMPTION

The power consumption varies depending on the operating mode, supply voltage and RF activity. All measurements are taken at the ambient temperature of 25°C. The current values represent the total module supply current of the VD1833 pin.

Power mode	Description	VD1833	Typ.	Unit
Active	KM0 & KM4 idle	3.3V	20	mA
	RF OFF KM4: 200MHz	1.8V	40	mA
Sleep	KM4 power-gating KM0 clock-gating	3.3V	75	µA
	RF OFF SRAM all retained	1.8V	89	µA
Deep-sleep	CPU OFF	3.3V	10	µA
	RF OFF Interrupt OFF SRAM 1KB retained	1.8V	10	µA

Table 5-5. Current consumption in low-power modes

RF Condition	Description	Typ.	Unit
TX	1T_MCS7 / BW40M (15dBm@2G)	206	mA
	1T_MCS7 / BW40M (15dBm@5G)	286	
	1T-Legacy OFDM54M (19dBm@2G)	262	
	1T-Legacy OFDM54M (19dBm@5G)	325	
RX	1R-MCS7 / BW40M (Pin = -60dBm@2G)	63	mA
	1R-MCS7 / BW40M (Pin = -60dBm@5G)	65	
	1R-Idle / BW40 @ 2G	50	
	1R-Idle / BW40 @ 5G	51	

Table 5-6. Current consumption for Wi-Fi® RF (3.3V)

RF Condition	Description	Typ.	Unit
TX	1T_MCS7 / BW40M (12dBm@2G)	205	mA
	1T_MCS7 / BW40M (11dBm@5G)	222	
	1T-Legacy OFDM54M (13dBm@2G)	201	
	1T-Legacy OFDM54M (12dBm@5G)	229	
RX	1R-MCS7 / BW40M (Pin = -60dBm@2G)	100	mA
	1R-MCS7 / BW40M (Pin = -60dBm@5G)	103	
	1R-Idle / BW40 @ 2G	80	
	1R-Idle / BW40 @ 5G	80	

Table 5-7. Current consumption for Wi-Fi® RF (1.8V)

5.5. MEMORY CONFIGURATION

The NU-76 is based on the RTL8720DF which include embedded Flash memory and on-chip SRAM. The memory is shared between the KM4 and KM0 processors by a multilayer AXI bus matrix.

Memory	Description	Size
Embedded Flash	Internal Flash	4 MB
KM4 SRAM	High-power main SRAM, accessible by KM4 & KM0	512 KB
KM0 SRAM	Low-power main SRAM, accessible by KM4 & KM0	64 KB
Retention SRAM	Data retention during deep-sleep mode	1 KB

Table 5-8. Memory configuration

6. RF CHARACTERISTICS

The RF Characteristics are measured at the antenna port with VD1833 = 3.3V and ambient temperature of 25°C. The range of target transmit power and center frequency can be configured by software. The module must operate in the center frequency range allocated by local regulatory authorities.

6.1. 2.4GHZ WI-FI® RADIO

Parameter	Operation mode		Specification (TBD)		Unit		
RF frequency range	802.11b/g/n		2412 – 2484		MHz		
Modulation	802.11b		CCK and DSSS				
	802.11g/n		OFDM				
Supported Data Rates	802.11b		1, 2, 5.5, 11		Mbit/s		
	802.11g		6, 9, 12, 18, 24, 36, 48, 54				
	802.11n		MCS0 – MCS7				
Supported Bandwidth	802.11n		20, 40		MHz		
Supported Guard Interval	802.11n		400, 800		ns		
			1.8V ±2dBm (TBD)	3.3V ±2dBm (TBD)			
Conducted Transmit Power (typical)	2.4 GHz CCK	1 Mbit/s	14	20	dBm		
		11 Mbit/s	12	18			
	2.4 GHz OFDM	6 Mbit/s	14	20			
		54 Mbit/s	11	17			
	2.4 GHz HT20 / HT40	MCS0	13	19			
		MCS7	10	16			
	Receiver Sensitivity (typical)	2.4 GHz CCK	1 Mbit/s	-98		-98	dBm
			11 Mbit/s	-91		-91	
2.4 GHz OFDM		6 Mbit/s	-96	-95			
		54 Mbit/s	-77	-77			
2.4 GHz HT20		MCS0	-95	-95			
		MCS7	-75	-75			
2.4 GHz HT40		MCS0	-93	-93			
		MCS7	-72	-72			

Table 6-1. 2.4 GHz Wi-Fi® radio characteristics

6.2. 5GHZ WI-FI® RADIO

Parameter	Operation mode		Specification (TBD)		Unit
RF frequency range	802.11a/n		5180 – 5825		MHz
Modulation	802.11a/n		OFDM		
Supported Data Rates	802.11a		6, 9, 12, 18, 24, 36, 48, 54		Mbit/s
	802.11n		MCS0 – MCS7		
Supported Bandwidth	802.11n		20, 40		MHz
Supported Guard Interval	802.11n		400, 800		ns
			1.8V ±2dBm (TBD)	3.3V ±2dBm (TBD)	
Conducted Transmit Power (typical)	5 GHz OFDM	6 Mbit/s		18	dBm
		54 Mbit/s		14	
	5 GHz HT20/HT40	MCS0		17	
		MCS7		13	
Receiver Sensitivity (typical)	5 GHz OFDM	6 Mbit/s	-94	-93	dBm
		54 Mbit/s	-76	-76	
	5 GHz HT20	MCS0	-94	-93	
		MCS7	-74	-74	
	5 GHz HT40	MCS0	-91	-91	
		MCS7	-71	-71	

Table 6-2. 5 GHz Wi-Fi® radio characteristics

6.3. BLUETOOTH® 5 LE RADIO

Parameter	Condition	Specification (TBD)	Unit
RF frequency range		2402 ~ 2480	MHz
Supported modes		Bluetooth LE	
Number of channels		40	
Modulation		GFSK	
Transmit power (typical, conducted)		4.5	dBm
Transmit power range		-10 ~ 10	dBm
Receiver sensitivity (typical)	LE 1M	-100.9	dBm
	LE 2M	-96.8	

Table 6-3. Bluetooth® LE radio characteristic

7. PHYSICAL DIMENSIONS

The physical dimensions of the NU-87 module are shown below. Dimensions are referenced from the top view of the module.

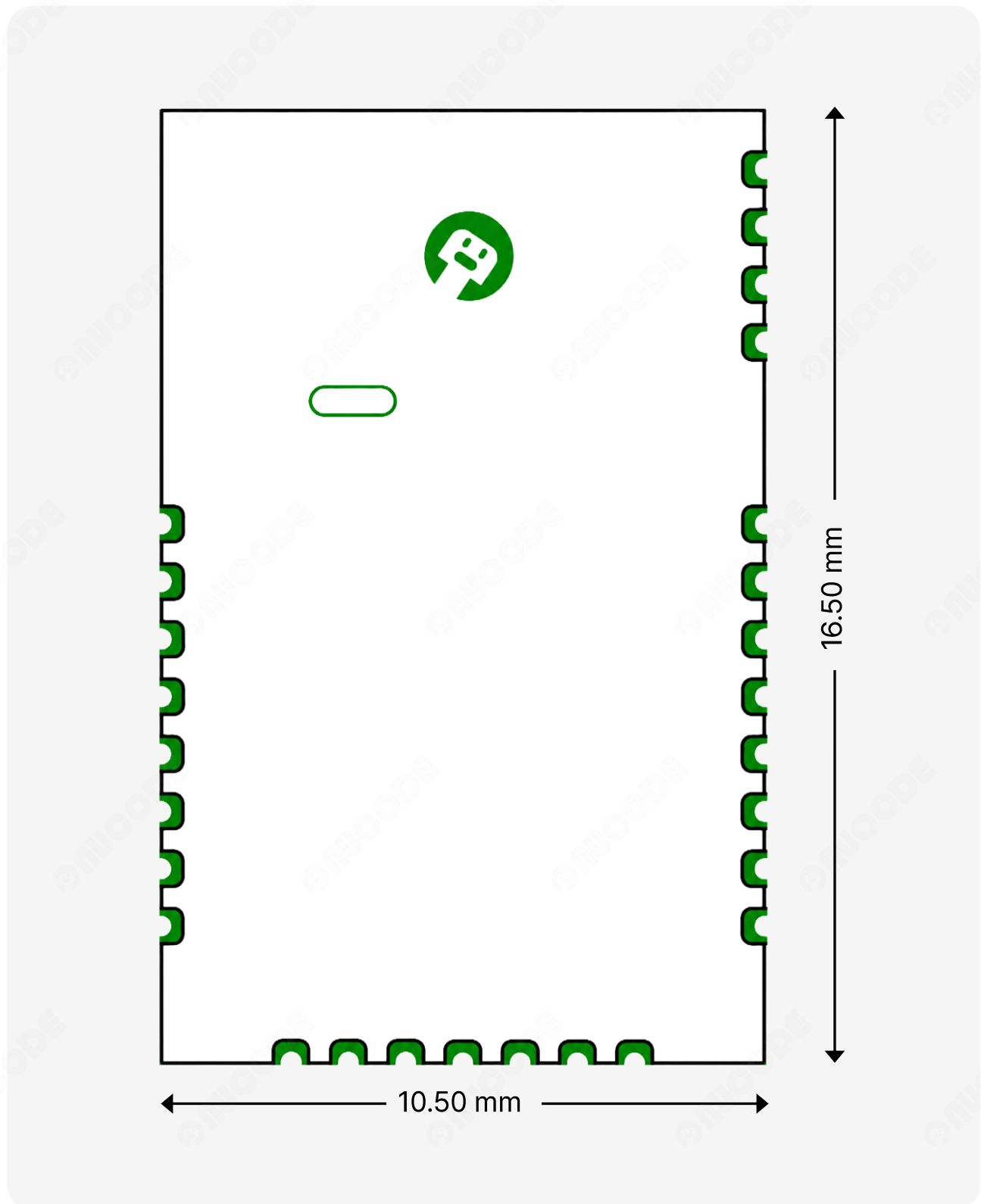


Figure 7-1. NU-87 Dimensions

8. PACKAGING SPECIFICATION

The packaging specification of the NU-87 are shown below.

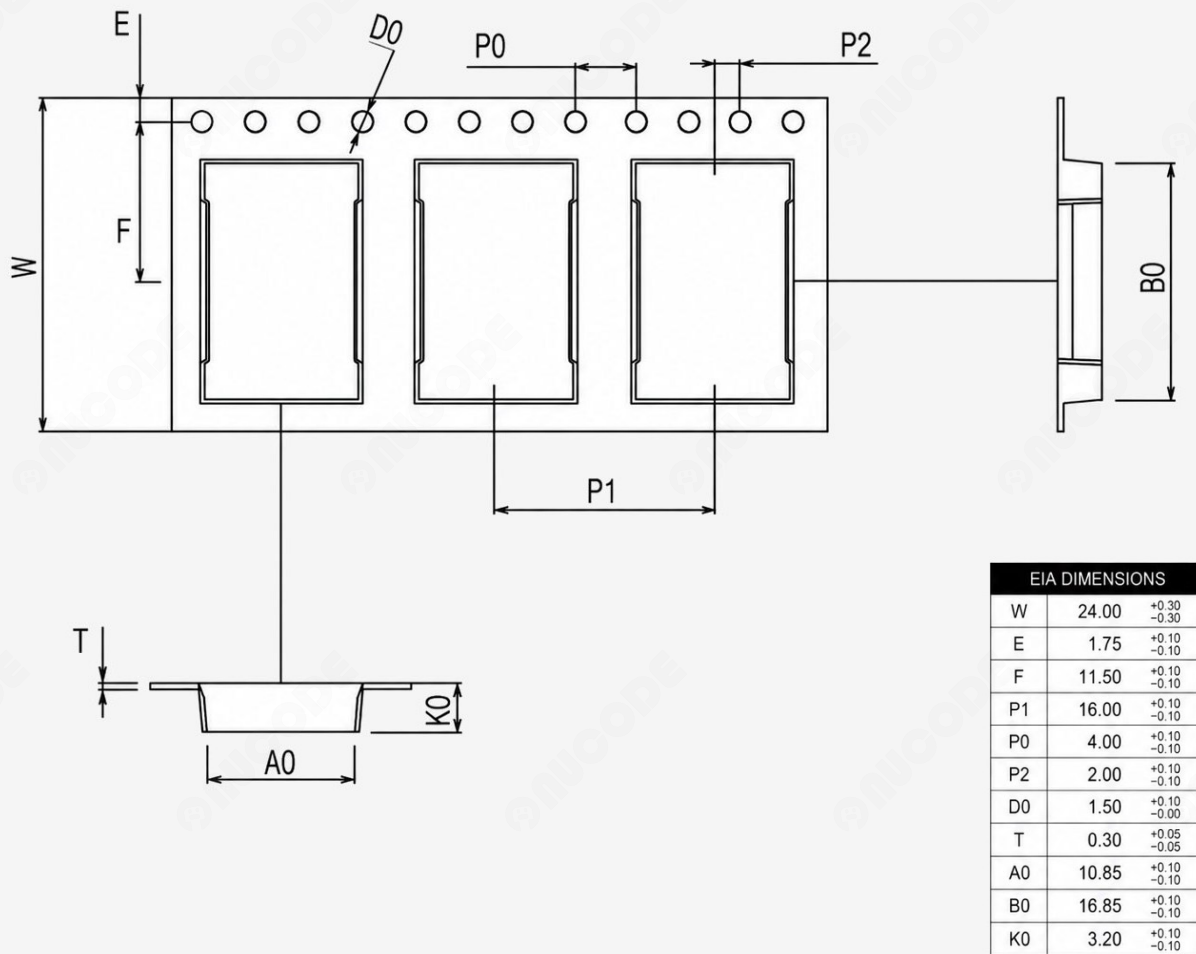


Figure 8-1. NU-87 Packaging Specification

ORDERING INFORMATION

Ordering code	Product
NCWB87R01VC	Module with Chip Antenna. Based on RTL8720DF
NCWB87R01VI	Module with antenna IPEX. Based on RTL8720DF

RELATED DOCUMENTATION

NUCODE Documentation archive: <https://nuworks.io/wiki/software>

I2C®: Registered trademark of NXP

I2S®: Registered trademark of NXP

Wi-Fi®: Registered trademark of Wi-Fi Alliance

Bluetooth®: Registered trademark of Bluetooth SIG, Inc.

REVISION HISTORY

Date	Version	Release Notes
2026-04-20	V1.0	First release

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